



Conditions & Logic

Introduction to Programming in Python

Lesson 05 · ReDI School of Digital Integration

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What did we learn last week?



Indexing & Slicing

```
name[0]  name[1:4]
```



String Methods

```
.upper()  .strip()  .replace()
```



f-strings

```
f"Hello, {name}!"
```

Today our programs learn to **make decisions!**

Today's Plan

Part 1 — if / else

- Comparing values
- Running code only when a condition is true
-  Exercise

Part 2 — elif

- Checking multiple options in order
-  Exercise



Break

Part 3 — and, or, not

- Combining conditions
-  Exercise

Part 4 — Nested conditions

PART 1 if / else

Making Decisions

Every day you make decisions based on conditions:

- **If** it is raining → take an umbrella
- **If** you are hungry → eat something
- **If** the light is red → stop

Python uses the same idea:

```
temperature = 32

if temperature > 30:
    print("It is hot today!")
```



The code inside the `if` block only runs when the condition is `True` .
If the condition is `False` , Python skips that block completely.

The if / else Structure

```
temperature = 32

if temperature > 30:
    print("It is hot today – wear light clothes!")
else:
    print("The weather is fine.")
```



⚠️ **Indentation matters!** The code inside `if` and `else` must be indented with 4 spaces (or one Tab). Python uses indentation to know what belongs inside the block.

```
# This is WRONG – Python cannot tell what belongs to the if
if temperature > 30:
print("hot")      # IndentationError!

# This is RIGHT
if temperature > 30:
    print("hot")  # 4 spaces inside
```



Comparison Operators

These operators compare two values and return `True` or `False` :

```
age = 20

print(age == 20)  # True   - equal to
print(age != 20)  # False  - not equal to
print(age > 18)   # True   - greater than
print(age < 18)   # False  - less than
print(age >= 20)  # True   - greater than or equal to
print(age <= 20)  # True   - less than or equal to
```



⚠ Common confusion:

`=` stores a value in a variable: `age = 20`

`==` checks if two values are equal: `age == 20`

Writing `if age = 20:` gives a **SyntaxError**.

Comparing Strings

You can compare strings too:

```
city = "Copenhagen"

if city == "Copenhagen":
    print("Welcome to Denmark!")

answer = input("Continue? yes or no: ")

if answer == "yes":
    print("Let's go!")
else:
    print("OK, stopping.")
```



⚠ String comparison is **case sensitive**.

"yes" , "Yes" , and "YES" are all different.

Use `answer.lower() == "yes"` to accept any case.

if / else — Full Example

```
score = int(input("What did you score? "))

if score ≥ 50:
    print("You passed – well done!")
else:
    print("You did not pass – try again next time.")
```



What happens step by step:

1. Python reads the score from the user
2. It checks: is `score ≥ 50` True or False?
3. If True → run the first block
4. If False → skip to `else` and run that block
5. Only one block ever runs

Exercise 1 — Weather Advice

Ask the user for today's temperature (as a number). Then print advice:

- If the temperature is above 25 → "Great weather for a walk!"
- Otherwise → "Maybe stay indoors today."

Bonus: also handle if it is below 0 — print "It is freezing — wrap up!"

```
temperature = int(input("What is the temperature today? "))  
  
# your code here
```



 8 minutes — try it yourself first!

Exercise 1 — Solution

```
temperature = int(input("What is the temperature today? "))

if temperature < 0:
    print("It is freezing – wrap up!")
elif temperature > 25:
    print("Great weather for a walk!")
else:
    print("Maybe stay indoors today.")
```



Test it:

```
What is the temperature today? 30    → Great weather for a walk!
What is the temperature today? 15    → Maybe stay indoors today.
What is the temperature today? -3    → It is freezing – wrap up!
```



PART 2 elif — Multiple Options

Choosing Between Many Options

`elif` means "else if". It lets you check several conditions in order. Python stops at the **first one that is True**:

```
score = 75

if score ≥ 90:
    print("Grade: A – Excellent!")
elif score ≥ 70:
    print("Grade: B – Good!")
elif score ≥ 50:
    print("Grade: C – Pass")
else:
    print("Grade: F – Not passed")

# Output: Grade: B – Good!
```



Think of `if / elif / else` like a series of gates.

Python checks them from top to bottom and goes through the **first gate that opens**. All the rest are skipped.

Order Matters!

Put the **most specific** condition first:

```
age = 70

# WRONG - age 70 matches the first condition, never reaches senior
if age > 18:
    print("Adult")
elif age > 65:
    print("Senior")    # this never runs!

# RIGHT - check the more specific case first
if age > 65:
    print("Senior")
elif age > 18:
    print("Adult")
```



⚠ When conditions overlap, always check the **smaller/more specific** range first.

elif — Another Example

```
hour = int(input("What hour is it? (0-23) "))

if hour < 12:
    print("Good morning!")
elif hour < 17:
    print("Good afternoon!")
elif hour < 21:
    print("Good evening!")
else:
    print("Good night!")
```



You can have as many `elif` blocks as you need.

The `else` at the end is optional — it catches everything that did not match.

Exercise 2 — Ticket Price

Write a program that asks for a person's age and prints the correct ticket price:

- Under 5 years old → **Free**
- 5 to 15 → **€5**
- 16 to 64 → **€12**
- 65 and over → **€7**

```
age = int(input("How old are you? "))  
  
# your code here
```



 8 minutes

Exercise 2 — Solution

```
age = int(input("How old are you? "))

if age < 5:
    print("Your ticket is free!")
elif age ≤ 15:
    print("Your ticket costs €5")
elif age ≤ 64:
    print("Your ticket costs €12")
else:
    print("Your ticket costs €7")
```



Test it:

```
How old are you? 3   → Your ticket is free!
How old are you? 12  → Your ticket costs €5
How old are you? 30  → Your ticket costs €12
How old are you? 70  → Your ticket costs €7
```





Break — 10 minutes

Back in 10 minutes

PART 3 and, or, not

Combining Conditions

Sometimes one condition is not enough. You can combine them:

```
age = 25
has_ticket = True

# AND – both must be True
if age >= 18 and has_ticket:
    print("You can enter!")

# OR – at least one must be True
if age >= 18 or has_ticket:
    print("Access allowed.")

# NOT – flips True to False, and False to True
is_closed = False

if not is_closed:
    print("The shop is open.")
```



and, or, not — How They Work

and — both must be True

A	B	Result
True	True	True
True	False	False
False	True	False
False	False	False

or — at least one True

A	B	Result
---	---	--------

Everyday Examples

```
day      = "Saturday"
weather = "sunny"

# Planning a picnic
if day == "Saturday" or day == "Sunday":
    if weather == "sunny":
        print("Perfect day for a picnic!")
    else:
        print("Weekend but not sunny – maybe next week.")
else:
    print("It is a weekday.")
```



```
# Checking a username input
name = input("Your name: ")

if name != "" and len(name) >= 2:
    print(f"Hello, {name}!")
else:
    print("Please enter a valid name (at least 2 characters).")
```



not — Practical Use

```
items = [] # empty shopping list

if not items:
    print("Your shopping list is empty.")
else:
    print(f"You have {len(items)} items.")
```



```
username = input("Username: ")
password = input("Password: ")

correct_user = "redi"
correct_pass = "python2026"

if username == correct_user and password == correct_pass:
    print("Welcome!")
else:
    print("Incorrect username or password.")
```



`not items` is True when the list is empty — this is a very common Python pattern.

Exercise 3 — Weekend Planner

Ask the user two questions:

1. What day is it? ("Saturday" or "Sunday" or a weekday name)
2. Is the weather good? ("yes" or "no")

Then print:

- Weekend **and** good weather → "Great day for outdoor plans!"
- Weekend **but** bad weather → "Staying in this weekend."
- Not a weekend → "It is a weekday – back to work!"

```
day      = input("What day is it? ")
weather = input("Is the weather good? (yes/no) ").lower()

# your code here
```



 8 minutes

Exercise 3 — Solution

```
day      = input("What day is it? ").strip().title()
weather = input("Is the weather good? (yes/no) ").lower()

is_weekend = day == "Saturday" or day == "Sunday"

if is_weekend and weather == "yes":
    print("Great day for outdoor plans!")
elif is_weekend and weather != "yes":
    print("Staying in this weekend.")
else:
    print("It is a weekday — back to work!")
```



Storing the weekend check in a variable (`is_weekend`) makes the code easier to read.

PART 4 Nested Conditions

Conditions Inside Conditions

You can put an `if` inside another `if`. This is called **nesting**:

```
has_ticket = True
age        = 15

if has_ticket:
    print("You have a ticket – checking age...")
    if age >= 18:
        print("Access to all areas!")
    else:
        print("Access to family areas only.")
else:
    print("Sorry – no ticket, no entry.")
```



Output:

```
You have a ticket – checking age ...
Access to family areas only.
```



When to Use Nesting

Use nested conditions when the second check only makes sense if the first one passed:

```
username = input("Username: ").strip()
password = input("Password: ").strip()

if username == "redi":
    if password == "python2026":
        print("Welcome!")
    else:
        print("Wrong password.")
else:
    print("Username not found.")
```



⚠ Each level of nesting adds 4 more spaces. Try not to nest more than 2 or 3 levels deep — it gets hard to read. Sometimes you can replace nesting with `and` :

```
if username == "redi" and password == "python2026":
    print("Welcome!")
```



Nesting vs and

Both approaches can work — choose the one that is clearer:

```
# Using nesting — good when the second check depends on the first
if has_ticket:
    if age ≥ 18:
        print("Full access")
    else:
        print("Limited access")
else:
    print("Buy a ticket first")

# Using and — good when you just need both to be True
if has_ticket and age ≥ 18:
    print("Full access")
elif has_ticket and age < 18:
    print("Limited access")
else:
    print("Buy a ticket first")
```



Exercise 4 — Coffee Order

Ask the user:

1. Do they want coffee? ("yes" or "no")
2. If yes — do they want milk? ("yes" or "no")

Print:

- Coffee with milk → "Here is your flat white!"
- Coffee without milk → "Here is your black coffee!"
- No coffee → "No coffee for you – enjoy your water!"

```
wants_coffee = input("Would you like coffee? (yes/no) ").lower()
```

```
# your code here
```



 8 minutes

Exercise 4 — Solution

```
wants_coffee = input("Would you like coffee? (yes/no) ").lower()

if wants_coffee == "yes":
    wants_milk = input("Would you like milk? (yes/no) ").lower()
    if wants_milk == "yes":
        print("Here is your flat white!")
    else:
        print("Here is your black coffee!")
else:
    print("No coffee for you – enjoy your water!")
```



Common Mistakes

Common Mistakes — Part 1

1. `=` instead of `==`

```
# WRONG - SyntaxError
if name = "Sara":

# RIGHT
if name == "Sara":
```



2. Forgetting `int()` with input

```
# WRONG - compares string to number!
age = input("Age: ")
if age >= 18:      # TypeError

# RIGHT
age = int(input("Age: "))
if age >= 18:
```



3. Wrong elif order

Common Mistakes — Part 2

5. Forgetting the colon :

```
# WRONG - SyntaxError
if score > 50
    print("Passed!")

# RIGHT
if score > 50:
    print("Passed!")
```



6. Wrong indentation breaks the logic

```
temperature = 35

if temperature > 30:
    print("It is hot.")
print("Drink water.") # This always runs - it is NOT inside the if!
else:
    print("Nice weather!") # SyntaxError - else has no if!
```





Mini Project — Smart Quiz

Build a 3-question quiz. Ask each question, check the answer, and keep a score. At the end, print the score and a message based on how they did.

```
score = 0

answer1 = input("What is the capital of France? ").strip().lower()
if answer1 == "paris":
    print("Correct!")
    score += 1
else:
    print("Wrong – the answer is Paris.")

answer2 = input("How many days are in a week? ").strip()
if answer2 == "7":
    print("Correct!")
    score += 1
else:
    print("Wrong – the answer is 7.")

answer3 = input("What colour do you get mixing red and blue? ").strip().lower()
if answer3 == "purple":
    print("Correct!")
    score += 1
else:
```



What We Learned Today

if / elif / else

- Run code only when a condition is True
- Use `elif` for multiple options
- `else` catches anything that did not match

Comparison operators

- `=` `≠` `>` `<` `≥` `≤`
- Always `=` for comparing, never `=`
- Works on numbers and strings

and, or, not

- `and` — both must be True
- `or` — at least one must be True
- `not` — flips True to False



Homework

Homework — Beginner (1–5)

1. **Grade Calculator** Ask for a score (0–100). Print:

- 90 and above → "Grade A"
- 75 to 89 → "Grade B"
- 60 to 74 → "Grade C"
- 50 to 59 → "Grade D"
- Below 50 → "Grade F"

2. **Even or Odd** Ask for a number. Print whether it is even or odd. Hint: use `%` — if `number % 2 == 0` it is even.

3. **Bigger Number** Ask for two numbers. Print which one is bigger, or say they are equal.

4. **Time of Day Greeting** Ask for the current hour (0–23). Print "Good morning!" (before 12), "Good afternoon!" (12–17), or "Good evening!" (after 17).

5. **Favourite Season** Ask for a month number (1–12). Print the season:

Homework — Intermediate (6–10)

6. Number Classifier Ask for a number. Print all that apply:

- Positive, negative, or zero
- Even or odd
- Divisible by 3 (yes or no)

7. BMI Calculator Ask for weight in kg and height in metres. Calculate BMI (`weight / height ** 2`). Print: "Underweight" (below 18.5), "Normal" (18.5–24.9), "Overweight" (25 or above).

8. Simple Calculator Ask for two numbers and an operation (`+` , `-` , `*` , `/`). Print the result. Handle division by zero — print an error message instead of crashing.

9. Vending Machine Ask for a product name and the amount of money inserted.

- Cola → €1.50, Water → €1.00, Juice → €2.00
- If money is enough → print the change
- If not enough → print how much more is needed
- If product not found → print an error message



Next Week — Loops

Right now your program runs each line once.

Next week you will make it **repeat things automatically**.



for loop

Do something for every item in a list



while loop

Keep going until a condition is met



range()

Count from any number to any number



How would you print "Hello" 100 times right now? Next week — one line of code.

Great work today! 🎉

Your programs can now think and make decisions



if / elif / else

Control the flow



and / or / not

Combine conditions



Nesting

Conditions inside conditions

See you next week! 🙌